

Question 1	
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Keypad (Pty) Ltd is a Cape Town based company which manufactures and markets a range of electronic security devices. A recent innovation, which has been fully researched in terms of the market potential for a new security device is being evaluated by the company. The financial accountant seems to have prepared schedules and lists of additional information, all of which is correct, but seems a little confused about how to use the information and perform the calculations required by management. You are asked to help.

1. Due to fast changing technology, the life of the new project is estimated at three years. CUL = 3
- ✓2. The cost of the marketing survey was R120,000.
- ✓3. The cost of new machinery required for the project will be R5 million.
- ✓4. None of the existing machinery will be sold or scrapped if the new machine is purchased. The new machinery will have a residual value of R1.1 million at the end of the project.
5. The new product is expected to cannibalise (*take away*) the pre-tax cash flows from another product of R15,000 per month over the next three years. R180k
- ✓6. Working capital of R400,000 will be required on commencement of the project.
7. Accounting depreciation calculated on a straight-line basis over the estimated life of the project whilst for tax purposes the SA Revenue Service will allow the section 12C wear and tear allowance on the 40%: 20%: 20%: 20% basis
8. Sales for 2015 to 2017 will be R5.5¹million, R6.5² million, and R7.5³ million respectively. The company maintains a consistent mark-up and cost of sales will be 40% of sales. Other costs (*including fixed costs*) will be 45% of sales. Included in the other costs are accounting depreciation and fixed costs allocated from Head Office amounting to R440,000 per annum for each of the next 3 years.
9. The financial accountant has consulted a textbook and started preparing the following schedule (*which is correct*) but did not know what to call the final result was (*marked ???*) or how to use it.

NPV

w1 New machine Recoupment

cost R5 m

Resid R1,1m

Tax value = R0

∴ Recoupment = R1,1m

Depr = $\frac{5m}{3} = 1,67m$

w2 old machine

- not being sold/scrapped

w3 Tax schedule

	1	2	3
Incremental CFs	5,5m	6,5m	7,5m
(Less) maintenance			
NCFs			
Recoupment (new)			1,1
(Less) Depreciation	(1,67m)	(1,67m)	(1,67m)
Taxable income			
Tax (28%)			

NPV schedule

	0	1	2	3
Initial ^(machinery) investment	(400k) 5m			
Marketing sunk cost ✓	-			
condabilize ^(op. cost) ✓		(180k) ✓	(180k)	(180k)
working cap	(400k)			
Incremental CFs / NOCI		2,5 5,5m	2,7 6,5	400 7,5
(Less) maintenance				
NCFs				
Residual				1,1m
Tax ✓				
NCFs				
Discount Factors @ 15%	1			
PV of NCFs				
NPV				595

∴ should be accepted because NPV > 0

wacc

	Years Ending 31 March		
	2015	2016	2017
	R'000	R'000	R'000
Sales Income	5 500	6 500	7 500
Less. Cost of sales	-2 200	-2 600	-3 000
Gross profit	3 300	3 900	4 500
Less. Other costs 45%	-2 475	-2 925	-3 375
Net profit before tax	825	975	1 125
Add. Depreciation (accounting depreciation)	1 300	1 300	1 300
Add. Allocated costs from Head office	440	440	440
What is this amount Called	2 565	2 715	2 865

Net operating cash inflows

10. The company tax rate is 28%.

11. Debt makes up 40% of the company's capital structure and the cost of debt before tax is 12.5%.
The cost of equity is 19 % and the Weighted Average Cost of Capital (WACC) is 15%.

$$K_d(1-t) = 9\%$$

REQUIRED:

1. Calculate the Net Present Value (NPV) of the project and advise the company on this basis whether it should undertake the project. (Note: Items omitted from the NPV calculation on the basis that they do not represent cash flows should be clearly identified).

Present your answer in an appropriate schedule working in thousands of rands (R'000). **Show all workings.**

Question 2

Mogidi Manufacturers is evaluating whether or not to replace its old and outdated existing machine that cost R12m three years ago with a remaining useful life of 1 year. A new technologically enhanced machine will cost R14m. The cost of transportation to the factory and transport and installation costs amounts to R1m. The existing machine can currently be sold for R2.9m with the proceeds received immediately.

The new machine is expected to have a useful life of 6 years and a residual value of R3m. However, the machine will only be used for 4 years. South African Revenue Services (SARS) allows a Section

Depreciation

12C wear and tear allowance of 40% in the first year and 20 % in the remaining three years. This was applicable to the old and the new machine.

The new machine will have fewer breakdowns, requires less servicing and is much faster resulting in an increase in annual sales as follows - R10m per annum in year one, R8m in year two, R7.5m in year three and year 4 the increase will only be R7m. The firm estimates that the value of closing and opening inventory will always be zero in each of the four years.

The production costs comprising cost of sales and fixed costs is estimated to be 70 % of the annual increased sales amount over the four years. The fixed costs include the annual depreciation on the new machine and company allocated costs of R1m per annum.

The firm would require working capital requirements in the form of raw materials inventory of R1m when the machine is ready and available for use and thereafter R2m in year 2, R1.5m in year 3. You may assume that the working capital is required at the beginning of the year and that all working capital will be recovered at the end of the project.

The firm's required cost of capital is estimated at 14 % and has a current company taxation rate of 28%.

REQUIRED:

1. Prepare a schedule using Net Present Value (NPV) Capital Budgeting technique to determine whether or not Mogidi Manufacturers should replace the existing machine or not using the. **State your conclusion. Please work in millions and to two decimal places. All workings must be clearly shown.**

w1 New machine Recoupment
 Cost R14m + 1m
 NPV Residual = R3m after 6 yrs
 Depr = 40%, 20%, 20%, 20%
 Tax value = R0
 Tax $\therefore$ Recoupment = R3m

w2 Old machine Recoupment
 Cost R12m
 NPV EUL = 4 yrs
 Selling @ 2,9m (Residual)
 Acc Depr = 80% $\times$ 12m = 9,6m
 Tax value = 12 - 9,6 = 2,4m
 Tax $\therefore$ Recoupment = R2,9m - R2,4m = 0,5m

w3 Tax schedule

	1	2	3	4	5	6
Incremental CFs	10	8	7,5	7		
CoS	(7)	(5,6)	(5,25)	(4,9)		

Recoup (new)						3
Recoup (old)						
(Less) depreciation						
Taxable income						
Tax (28%/-)						

NPV schedule

	0	1	2	3	4	5	6
Initial investment	(14)						
working capital ✓	(1) ✓	(2) ✓	(1,5) ✓		1,5		4,5
Additional cost	(1)						
Incremental CFs ✓		10	8	7,5	7 ✓		
(Less) costs w3		(4)	(2,6)	(2,25)	(1,9)		
Residual (old) ✓	(2,9)	2,9 ✓					
Residual					3		3
Tax							
NCFs							
Discount factors @ 14% ✓							
PV of NCFs ✓							
Sum of							
NPV							2,56

$\therefore$ replace

w/s
 Production costs 70% Sales (1) (5,6) (5,25) (4,90)
 + Depreciation
 + Allocated costs
 Product Cos (4) (2,6) (2,25) (1,9)

Solution

Question 1	0	1	2	3
	R'000	R'000	R'000	R'000
Machinery required	-5 000			
Residual value-new machine				1 100
Mkt research - sunk cost				
Net Operating Cash Inflows (Given)		2 565	2 715	2 865
Opportunity cost – products (R15 000 x 12)		-180	-180	-180
Working capital	-400			400
Taxation		-107.8	-429.8	-499.8
Total	-5 400	2 277	2 105	3 685
PV Interest Factor at 15 %		x0.8696	x0.7561	x0.6575
PV of cash flows		1 980	1 592	2 423
Sum of Cash flows	5995			
NPV	595			

Decision: Project should be accepted

Taxation Schedule			
	1	2	3
Net Operating Cash inflows (given)	2 565	2 715	2 865
Opportunity costs (15 000 x 12)	-180	-180	-180
Section 12 C wear and tear (40%, 20%, 20%, 20%)	-2 000	-1 000	-1 000
Recoupment			100
Taxable income	385.0	1 535	1 785
Taxation @28%	-107.8	-429.8	-499.8

Question 2 solution below

Question 2

	0	1	2	3	4
	Rm	Rm	Rm	Rm	Rm
Initial cost	-14				
Additional cost	-1				
Annual cash flows		10	8	7.5	7
Less. Production costs (COS) W1		-4	-2.6	-2.25	-1.90
Residual value (old)	2.9				
Working capital	-1.0	-1.0	0.5		1.5
Residual value (new)					3
Taxation		-0.81	-0.67	-0.63	-1.43
Totals	-13.1	4.19	5.23	4.62	8.17
Discount factors@14%		0.8772	0.7695	0.6750	0.5921
PVs of cash flows		3.68	4.02	3.12	4.84
Sum of PVs	15.66				
NPV	2.56				

Taxation schedule

	1	2	3	4
	Rm	Rm	Rm	Rm
Annual cash flows	10	8	7.5	7
Less. Production costs (COS) W1	-4	-2.6	-2.25	-1.90
Section 12 C wear and tear	-6	-3	-3	-3
Lost section 12 C allowance	2.4			
Recoupment (old)	0.5			
Recoupment new				3
Taxable income	2.9	2.40	2.25	5.10

Taxation @28%	0.81	0.67	0.63	1.43
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Workings (W1)

Production costs 70% of sales	-7	-5.6	-5.25	-4.90
Depreciation	2	2	2	2
Allocated costs	1	1	1	1
Product cost of sales	-4	-2.6	-2.25	-1.90

W2 depreciation New $(15-3)/6 = 2$

At cost (old) W3	12
Acc. Dep sec 12 c	9.6
Tax value old	2.4

Recoupment old $(2.9 - 2.4) = 0.5$

W4

At cost new	15
Acc. Dep sec 12 C	-15
Tax value	0

Recoupment new $(3-0) = 3$